

SI26_P06: Career Fair Eligibility Shortlist

AI-Assisted Coding Interview Problem

Problem Statement

Build a compact Career Fair Eligibility Shortlist for a university placement team. It should compare one student's academic profile and skills with a fixed local list of career-fair roles, then show which roles the student is eligible for and every failed rule for each ineligible role.

Use one attractive primary screen or report with an editable student profile, the five fixed role requirements, an Evaluate action, a result list, eligible/ineligible counts, a validation message, and sample/reset controls. You may use a browser, desktop or mobile tools, a spreadsheet or notebook, or a CLI that produces a clear visual or tabular report. No ranking, application tracking, sensitive personal attributes, interview scheduling, booking, account, backend, network service, or external job feed is required.

Load this built-in student profile:

Field	Value
Branch	CSE
CGPA	8.1
Graduation year	2027
Active backlogs	1
Skills	Git, Python, SQL

Use these fixed local roles:

Role ID	Role	Allowed branches	Minimum CGPA	Allowed graduation years	Maximum active backlogs	Required skills
CF01	Data Operations Intern	CSE, IT	7.5	2027	1	Python, SQL
CF02	QA Automation Intern	CSE, ECE, IT	7.0	2027, 2028	1	Git
CF03	Embedded Systems Intern	ECE, EEE	7.5	2027	1	Git

Role ID	Role	Allowed branches	Minimum CGPA	Allowed graduation years	Maximum active backlogs	Required skills
CF04	Machine Learning Intern	CSE, IT	8.5	2027	1	Python
CF05	Platform Engineering Intern	CSE, ECE	7.0	2026	0	Docker, Git

The built-in result has two eligible roles: CF01 and CF02. CF03 is ineligible because the branch is not allowed, CF04 is ineligible because the CGPA is below its minimum, and CF05 is ineligible because the graduation year is not allowed, the active-backlog limit is exceeded, and Docker is missing.

Contracts

- Trim surrounding spaces from branch and skill values. Compare branches and skills case-insensitively after trimming; do not infer branch aliases or related skills. Split an entered skill list on commas, ignore empty pieces, and collapse duplicates.
- CGPA must be a finite number from 0 through 10, inclusive. Graduation year must be a whole number from 2000 through 2100, and active backlogs must be a whole number greater than or equal to 0. A blank branch reports INVALID_BRANCH; invalid numeric fields report INVALID_CGPA, INVALID_GRADUATION_YEAR, or INVALID_BACKLOG_COUNT.
- A role is ELIGIBLE only when the branch is in its allowed set, CGPA is greater than or equal to its minimum, graduation year is in its allowed set, active backlogs are less than or equal to its maximum, and every required skill is present.
- Evaluate every rule independently. An ineligible role lists every failed rule exactly once using this order: BRANCH_NOT_ALLOWED, CGPA_BELOW_MINIMUM, GRADUATION_YEAR_NOT_ALLOWED, TOO_MANY_ACTIVE_BACKLOGS, then one MISSING_SKILL: <skill> reason per missing skill in case-insensitive alphabetical order. An eligible role shows no failure reasons.
- Show ELIGIBLE roles first and INELIGIBLE roles second. Within each status, sort by role title in case-insensitive ascending order, then by role ID in ascending order.
- Any invalid student-profile field clears earlier role results and eligible/ineligible counts instead of leaving stale values visible. Reset restores the complete built-in profile and role list.

Acceptance criteria

- **Required:** Load the built-in profile in one action and show CF01 and CF02 as ELIGIBLE, followed by CF03, CF04, and CF05 as INELIGIBLE, with counts of 2 eligible and 3 ineligible.
- **Required:** Show only BRANCH_NOT_ALLOWED for CF03, only CGPA_BELOW_MINIMUM for CF04, and exactly GRADUATION_YEAR_NOT_ALLOWED, TOO_MANY_ACTIVE_BACKLOGS, and MISSING_SKILL: Docker in that order for CF05.
- **Required:** Change only the CGPA to 8.5; CF04 must become eligible, and the eligible group must appear as CF01, CF04, CF02 by title with counts of 3 eligible and 2 ineligible.
- **Required:** Enter CGPA 10.5, report INVALID_CGPA, and clear all role results and counts.
- **Required:** Keep the profile, role requirements, ordered results, counts, validation message, and sample/reset actions synchronized, and include focused checks for the built-in result, the 8.5 boundary, complete failure reasons, ordering, and invalid CGPA.
- **Optional:** Add a compact role-card view driven by the same eligibility results.

Use AI coding assistants. Before implementation, create a short plan with 3–5 ordered steps and useful checkpoints. Be prepared to present that plan, explain any changes you made to it, share relevant prompts, summarize your design, and show test evidence such as tests, screenshots, or output samples.

How You'll Be Evaluated

- **Planning and Solution Presentation:** Present your 3–5-step implementation plan, explain how the work followed or changed that plan, and demonstrate the working solution with clear explanations
- **AI Prompting Strategy:** Show the prompts you used to translate this problem statement into technical specifications for AI assistants
- **Design Constraints and Technology Choices:** Explain the constraints you provided to AI regarding design patterns, technology stack, and architectural decisions
- **AI-Influenced Decision Making:** Discuss trade-offs, assumptions, and how AI recommendations influenced your choices for components, data structures, and implementation approaches
- **Testing and Validation:** Demonstrate how you tested the application covering both typical usage scenarios and edge cases
- **Live Modification Capability:** Be prepared to implement one small modification, and possibly a second if time permits, using AI assistance; keep your development environment ready for focused changes and verification